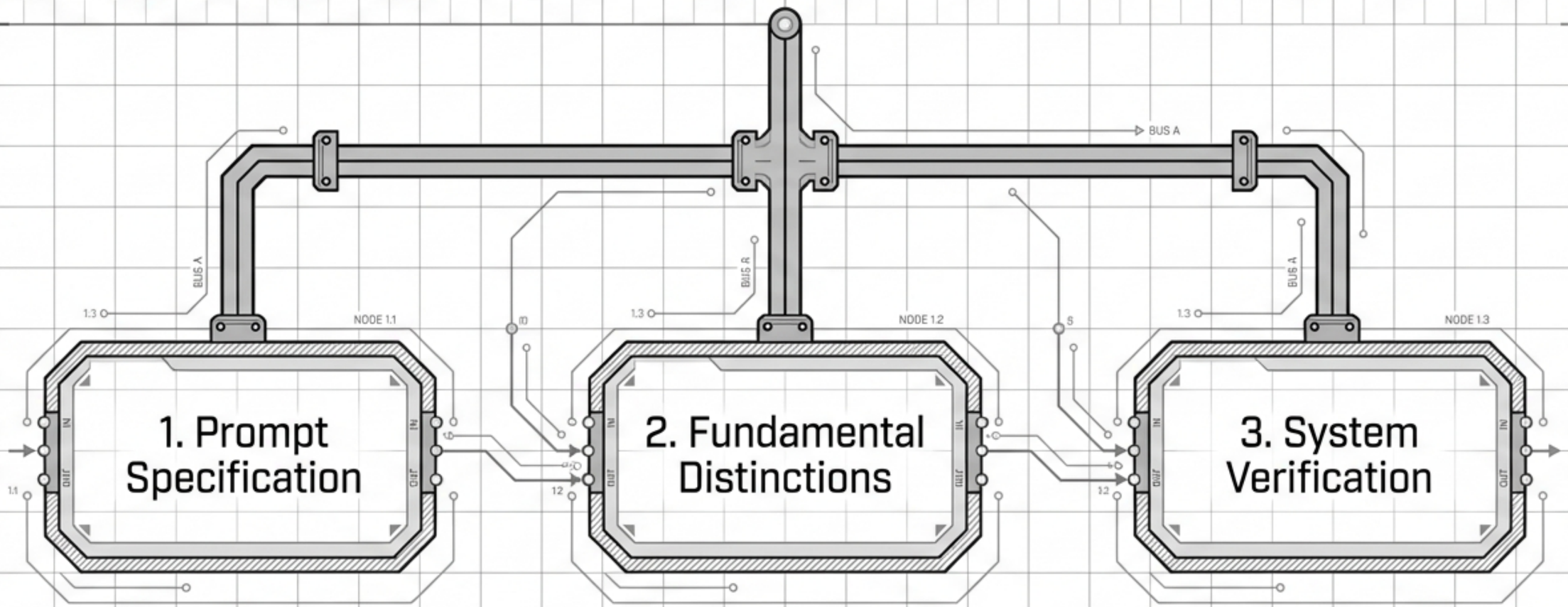
MODULE 14, LESSON 1

AI 101 FOR MECHANICAL ENGINEERS: WRITE AN ENGINEERING PROMPT

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Course Outline



Architecture Overview V1.0

System Flowchart

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1. Prompt Specification
2. Fundamental Distinctions
3. System Verification

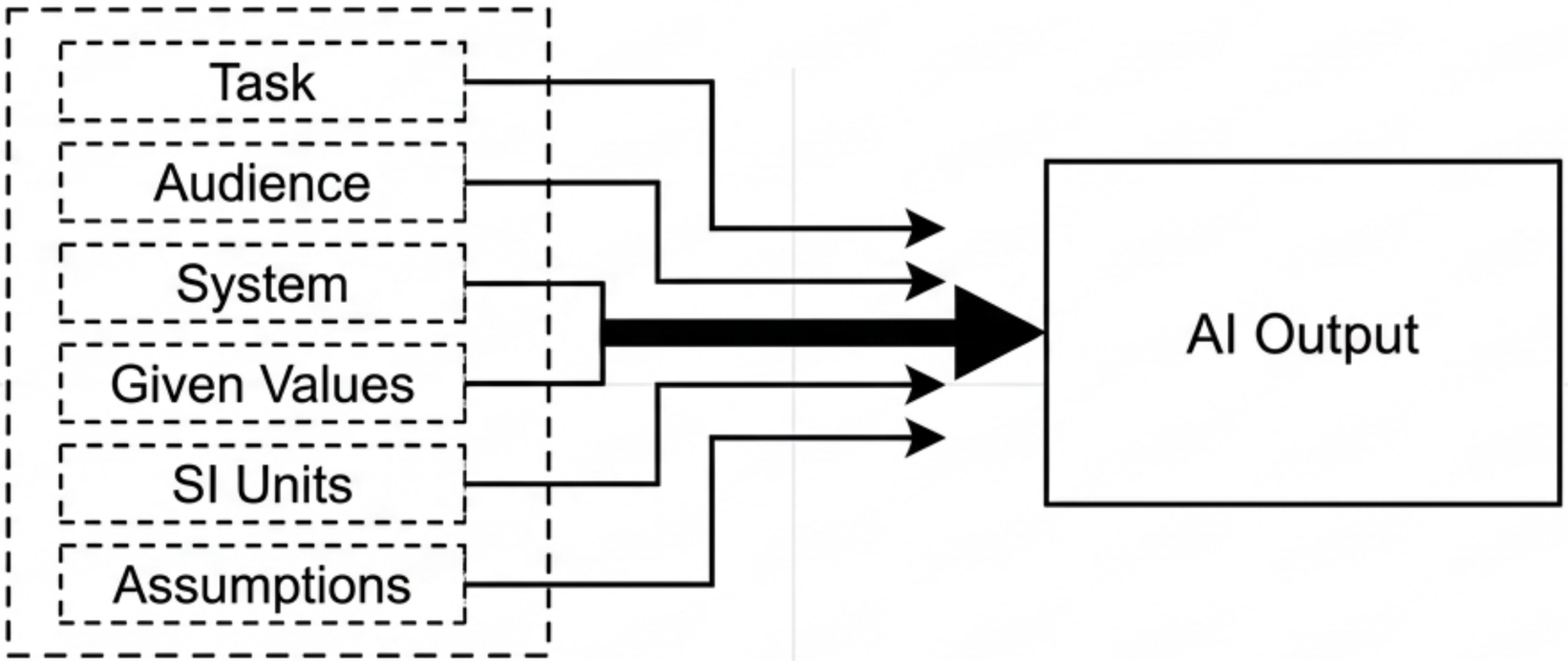
1. Prompt Specification

Establishing the inputs, definitions, and mathematical constraints for interacting with an AI model.

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1. Prompt Specification
2. Fundamental Distinctions
3. System Verification

The Engineering Prompt Architecture



State task, audience, and system explicitly.

Define values, SI units, and assumptions.

Request explicit distinction of verified facts.

Ask model to identify missing information.

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1. Prompt Specification
2. Fundamental Distinctions
3. System Verification

Prompting Typology Matrix

Method	Context Provided	Demonstration Examples	Functional Outcome
Zero-shot prompting	Task definition only	None supplied	Direct task execution
Few-shot prompting	Task definition plus format context	Desired output formats supplied	Constrained task execution

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1. Prompt Specification
2. Fundamental Distinctions
3. System Verification

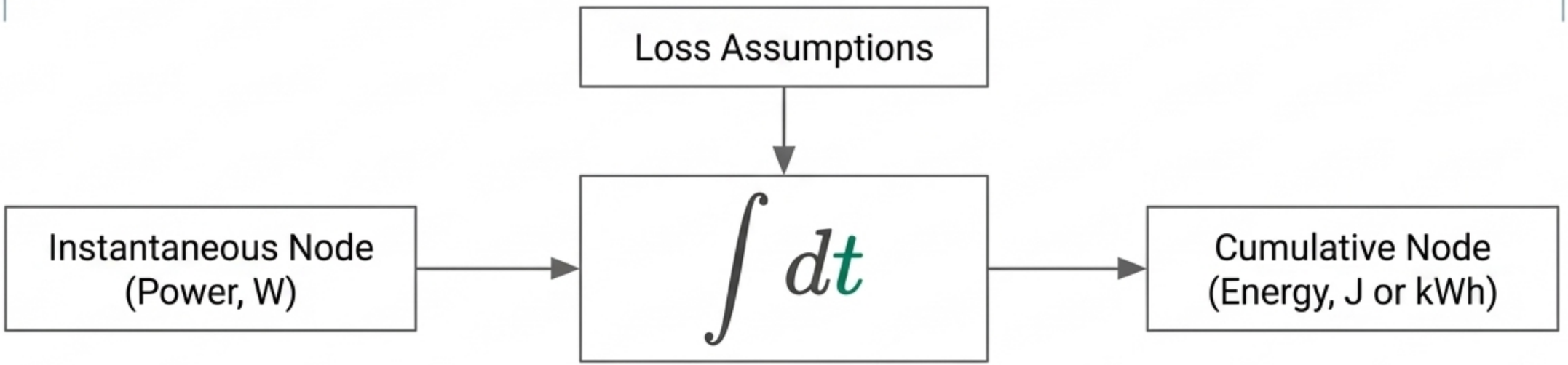
2. Fundamental Distinctions

Clarifying physical principles and vocabulary required to construct valid mathematical constraints.

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Power vs. Energy Integrator

1. Prompt Specification
2. Fundamental Distinctions
3. System Verification



Instantaneous irradiance cannot yield daily energy.

Explicit time information is fundamentally required.

Relevant system losses must be integrated.

Independent mathematical validation is strictly necessary.

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1. Prompt Specification
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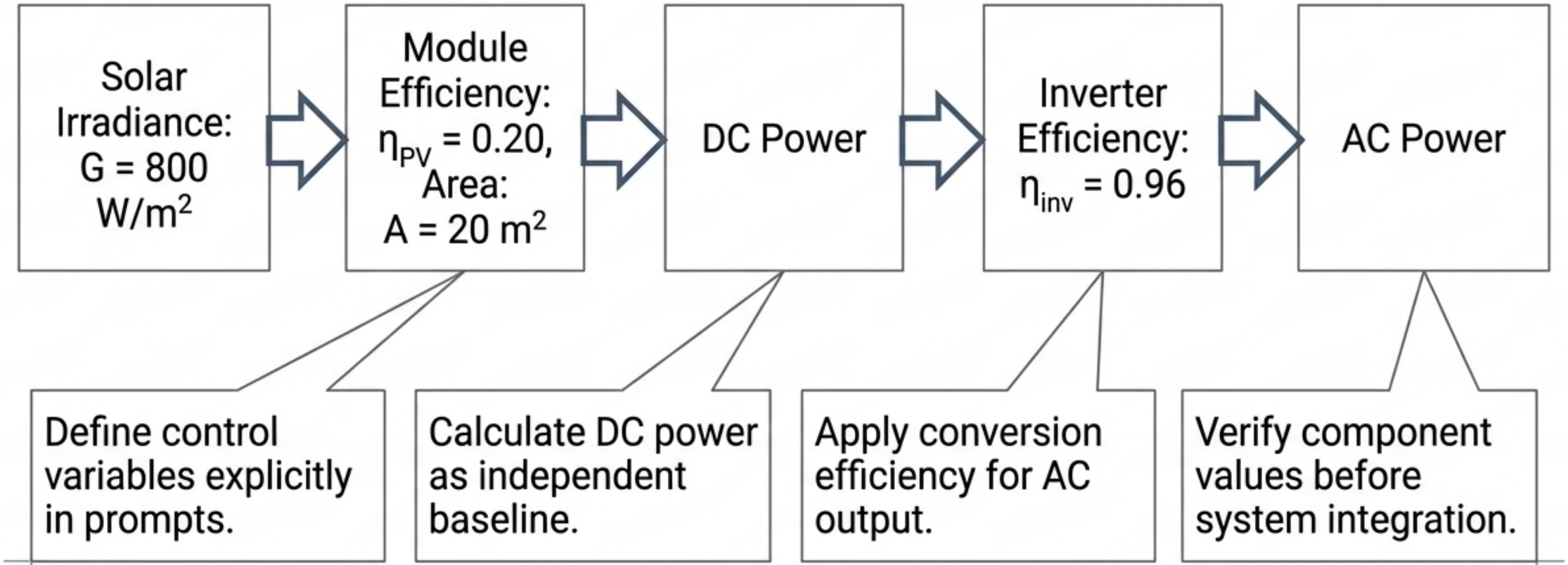
3. System Verification

Executing mathematical calculations, establishing reference parameters, and performing independent checks.

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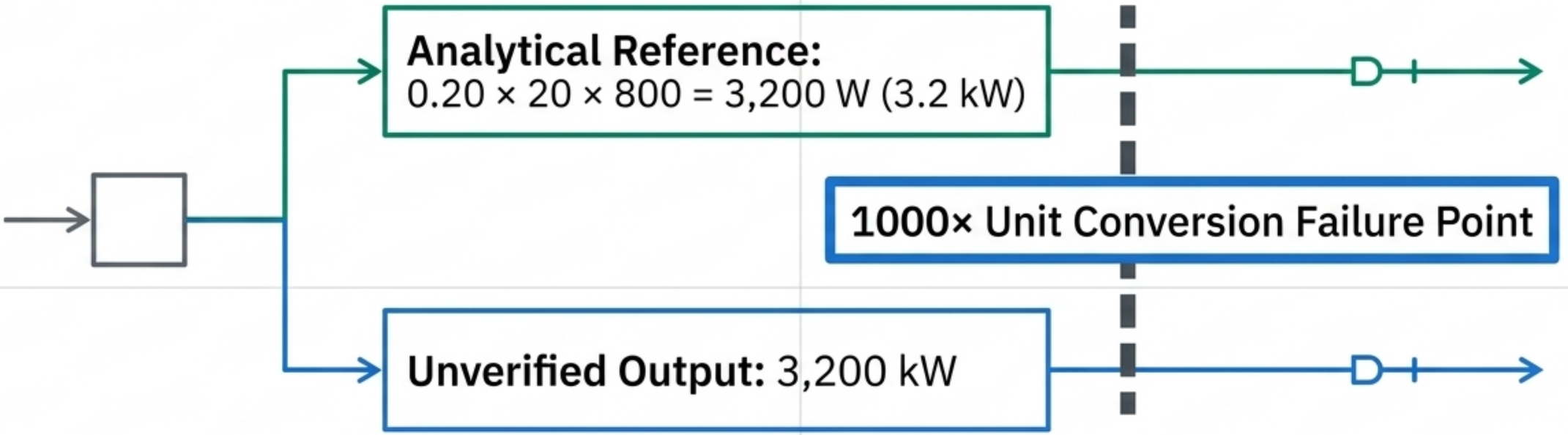
1. Prompt Specification
2. Fundamental Distinctions
3. System Verification

Photovoltaic Mathematical Sequence



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System Error Detection Schematic



Unverified AI outputs pose systemic risks.

Source material demonstrates a factor-of-1,000 error.

Draft outputs require independent mathematical calculation.

Test AI logic on known reference cases.

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1. Prompt Specification
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Independent Verification Matrix

Target	Equation	Verified Value
DC Power	$P_{\text{DC}} = \eta \cdot A \cdot G$	3,200 W
AC Power	$P_{\text{AC}} = P_{\text{DC}} \cdot \eta_{\text{inv}}$	3,072 W
Daily Energy (Constant 2h)	$E = \int P_{\text{AC}}\, dt$	6.144 kWh

Compare AI outputs against analytical references.

Confirm instantaneous DC and AC calculations.

Validate energy through explicit time integration.

Revise prompt based on detected errors.

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1. Prompt Specification
2. Fundamental Distinctions
3. System Verification

Key Points

1. Prompt Specification	Formulate prompts as explicit system inputs with mathematically defined constraints.
2. Fundamental Distinctions	Recognize time profiles and system losses as mandatory integration variables for energy.
3. System Verification	Mathematically validate all AI-generated outputs using independent reference equations.

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